

Submission Summary

Conference Name

International Conference on Sustainable Technology

Track Name

Track 5: Green Computing for Sustainability

Paper ID

260

Paper Title

Sustainable Healthcare Through Early Autism Spectrum Disorder Detection: A Multi-Model Machine Learning Framework for Survey-Based Assessment and Classification

Abstract

This study explored advancements in the early detection of Autism Spectrum Disorder (ASD) through an integrated approach combining survey-based methods with machine learning (ML) techniques. Our methodology implemented a comprehensive framework utilizing multiple algorithmic paradigms, including Random Forest (RF), XGBoost (XGB), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN). Data was collected through structured questionnaires administered during awareness seminars and targeted dissemination to parents in diverse settings, ensuring broad demographic representation. The framework incorporated advanced preprocessing techniques, including Synthetic Minority Over-sampling Technique (SMOTE) for class imbalance mitigation, feature engineering optimization, and strategic imputation methods to enhance dataset quality. Experimental results demonstrated exceptional performance across models, with RF and SVM achieving 98% accuracy, while XGB and KNN achieved 95% and 93% accuracy respectively. Cross-validation analysis (0.863 ± 0.061) confirmed robust generalization capabilities. Feature importance analysis through RF and XGB revealed critical behavioral markers, with communication abilities (importance: 1.222) and social interaction patterns (0.112) emerging as primary diagnostic indicators. The models demonstrated strong class-wise performance, with RF achieving precision/recall/F1 scores of 1.00/0.96/0.98 for non-ASD and 0.95/1.00/0.97 for ASD cases. These findings underscore the effectiveness of integrating machine learning techniques with survey-based diagnostic tools, establishing a clinically interpretable framework that enhances the accuracy and efficiency of early ASD detection while maintaining diagnostic reliability.

Created

1/11/2025, 11:15:36 AM

Last Modified

1/11/2025, 11:15:36 AM

Authors

Shilpa R (Vidyavardhaka College of engineering) <shilpar.ise@vvce.ac.in>

SriKrishna S Kashyap (Liverpool John Moores University) <uplskash@ljmu.ac.uk>

Vinay B K (Vidyavardhaka College of engineering) <Vinaybk@vvce.ac.in>

Rajendra A B (Vidyavardhaka College of engineering) <abrajendra@vvce.ac.in>

Vikram Athreya (The National institution of Engineering) <va.mech@nie.ac.in>

Submission Files

ASD ICST_NIE_.docx (288.3 Kb, 1/11/2025, 11:15:21 AM)

ASD ICST_NIE_.pdf (199.9 Kb, 1/11/2025, 11:15:21 AM)

Advancements in Early Autism Spectrum Disorder Detection: A Comparative Study of Survey-Based Methods Involving Machine Learning, and Deep Learning Models

Introduction

A lifelong disorder that impacts brain development, autism spectrum disorder (ASD) is characterized by repetitive behaviors, limited interests, and persistent difficulties with social communication that typically begin in early childhood [1], [2]. According to recent research, the number of ASD cases worldwide is increasing due to improved diagnosis techniques, more awareness, and potential genetic and environmental factors [1]. Globally, the rates vary, with a median of 6.2 cases per 1,000 children in Europe and 18.5 cases per 1,000 children in the United States [3]. According to a 2018 study conducted in India, the prevalence of ASD is roughly 10 per 1,000 children, or 1 in 100 [1], [3], and [4]. Globally, this increase in prevalence significantly impacts social support, education, and healthcare. Early diagnosis and treatment are essential for better long-term results for those with ASD [5]. Screening tools help spot early signs of ASD, allowing quick access to vital therapies and support [6]. Studies show that early, strong interventions focusing on behavioral and developmental methods can improve social skills, communication, and behavior [5], [6]. Recently, machine learning and deep learning methods have shown good outcomes in detecting and diagnosing ASD early. These techniques are key in analyzing questionnaire data for ASD identification. Questionnaires for early autism detection are non-invasive, easy to administer, and cost-effective, making them ideal for widespread use. They minimize discomfort and can be completed by caregivers, offering insights into a child's behavior. These tools enable early identification of developmental concerns without requiring complex medical procedures.

Early detection facilitates timely intervention, improving outcomes and reducing the burden on families [7]. The integration of AI and ML methods into questionnaire-based approaches for early autism detection has significantly enhanced their accuracy and efficiency. These technologies analyze patterns in responses, enabling the identification of subtle behavioral markers that might be missed by traditional methods. AI/ML tools also allow for personalized assessments, adapting questionnaires to the individual's unique characteristics. By processing large datasets, these methods improve predictive accuracy and reduce the likelihood of false positives. This advancement ensures earlier and more precise identification, enabling timely

interventions and better developmental outcomes for children at risk [11]. Support Vector Machines (SVM) can classify individuals based on their answers, helping to tell ASD apart from non-ASD cases [8]. Artificial Neural Networks (ANN) find patterns in questionnaire data, improving how ASD is detected [9]. Also, Decision Trees and Ensemble Methods like Random Forests (RF) and Gradient Boosting Machines find important features related to ASD [10]. Moreover, Deep Learning Models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), pick up detailed patterns from questionnaire data, which helps in better detection and scaling [11]. A key benefit of AI algorithms in early ASD detection is their quick and precise ability to find patterns in large datasets that might not be easy for human experts to see [12].

Literature Survey:

Abbas et al. [13] propose a machine learning-based autism detection system combining parent questionnaires and home video screenings, achieving notable improvements over traditional tools like M-CHAT and CBCL. The parent questionnaire classifier achieved an AUC of 0.991, sensitivity of 1.000, and specificity of 0.939 during training. Clinical validation of the combined approach yielded an AUC of 0.72, with specificity improving to 0.67 at 80% sensitivity. Including up to 25% inconclusive results enhanced accuracy while maintaining reliable predictions. This hybrid model performs better than existing methods, offering a low-cost, efficient alternative for early autism screening. Thabtah & Peebles [14] evaluated and introduced a rule-based machine learning algorithm, **Rules-Machine Learning (RML)**, utilizing thresholds for Minimum Frequency (Min_Freq) and Rule Strength (R_S) to generate efficient, non-redundant rule sets. RML outperformed models like C4.5, CART, PRISM, and ensemble techniques (Boosting, Bagging) across child, adolescent, and adult datasets. It achieved accuracies of 96.59%, 93.70%, and 94.20% for adults, adolescents, and children, respectively, with sensitivity and specificity exceeding 93% in all cases. Abdullah et al. [15] evaluated machine learning algorithms for ASD classification using a UCI dataset of 704 instances and 21 attributes. After preprocessing, five classifiers were applied and assessed via 5-fold cross-validation. Random Forest achieved the best performance (97.58% accuracy, 100% sensitivity, 96.66% specificity, AUC 1.00), followed by Decision Tree (96.87% accuracy, 98.21% sensitivity, 96.25% specificity, AUC 0.99). SVM showed strong results

(90.19% accuracy, AUC 0.94), while Naive Bayes (83.94% accuracy) and KNN (78.83% accuracy) performed lower. Random Forest emerged as the top classifier.

Jacob et al. [16] proposed an automated machine learning (AutoML) framework using JADBio for autism detection and feature signature discovery across different age groups. The framework achieved high accuracy: toddlers (96.1%, MCC 0.88), children (91.8%, MCC 0.81), adolescents (95.4%, MCC 0.89), and adults (97.2%, MCC 0.94). AutoML outperformed traditional machine learning models (evaluated via Orange) and identified influential non-clinical markers for autism classification. The study highlights AutoML's potential for accurate detection and efficient feature selection across diverse populations. Doe et al. [17] use the Autism Quotient (AQ-10) and individual characteristics to diagnose ASD with machine learning. Models include ANN, SVM, and RF, trained on a dataset of 701 samples. The ANN has three hidden layers with ReLU activation, achieving 90.34% accuracy without feature selection and 100% with RFE. SVM consistently achieves 100% accuracy, and RF improves from 95.72% to 100% with RFE. Feature selection using RFE significantly enhances model performance, especially in handling missing values. The algorithms used in the study by Neeharika et al. [18] for autism detection using questionnaires are Decision Tree, KNN, SVM, Logistic Regression, and Naive Bayes. The accuracy achieved by each algorithm is as follows, Decision Tree (Accuracy: 97.61, F1-Score: 97.75), SVM (Accuracy: 97.26, F1-Score: 97.41), KNN (Accuracy: 95.89, F1-Score: 95.83), Logistic Regression (Accuracy: 95.55, F1-Score: 95.45), and Naive Bayes (Accuracy: 96.23, F1-Score: 96.19).

The study by Bala et al. [19] utilizes machine learning for autism detection, combining questionnaire data with home video screenings. Reported accuracies include Logistic Regression (99.82%), SVM (99.73%), and Naive Bayes (91.74%). While CNNs aren't explicitly detailed in this study, related works report accuracies between 96.88% and 99.53%. Shahamiri et al. [20] utilize the AQ-10 and Q-CHAT-10 questionnaires for autism screening across age groups (toddlers to adults). Their primary algorithm, a Convolutional Neural Network (CNN), achieved an accuracy of 97.95%. The study also compared the CNN to Decision Trees, Rule Induction, and Bayes Net, noting the CNN outperformed them on metrics like accuracy, sensitivity, specificity, and true positive rates. The study by Abbas et al. [21] employed machine learning, primarily a Random Forest classifier, to enhance autism detection through parental questionnaires using ADI-R data. The "With inconclusive allowance" variant achieved an AUC of 0.991, sensitivity of 1.000, and specificity of 0.939, while the "Aggregate

features" variant achieved an AUC of 0.972. These results showed significant accuracy improvements over traditional tools like M-CHAT and CBCL across age groups.

The study by Wall et al. [22] used the **ADOS Module 1**, assessing key autism domains like language, social interaction, and play. Machine learning reduced the questionnaire to 8 critical items, maintaining diagnostic accuracy. Key algorithms and their accuracies include **ADTree** (100%), **FT** (100%), **J48** (99.4%), and **Random Tree** (97.8%). **ADTree** was chosen for its 100% sensitivity and specificity using only 8 items. The streamlined version enables faster, more reliable autism screening. This work proposes [23] that a Hyperparameter tuning was applied to SVM, kNN, RF, and ANN classifiers to classify ASD and typical development (TD). Data preprocessing included One-Hot encoding, kNN imputation, and MinMaxScaler normalization. The Random Forest (RF) classifier achieved 100% accuracy across all age groups without overfitting, outperforming methods like DNN and CNN, which reached up to 98% accuracy. RF demonstrated superior generalization and simplicity, making it ideal for real-time ASD detection. The study emphasizes its effectiveness during clinical trials. The model uses [24] a **Self-Organizing Map (SOM)** for clustering data based on traits like communication and social behavior, followed by **classification algorithms** (e.g., kNN, SVM, Random Forest). Hyperparameter tuning adjusts parameters like grid size for SOM and k for kNN. **Performance metrics** include accuracy, precision, recall, F1-score, and ROC-AUC. The model showed improved accuracy, precision, and recall when using the refined dataset from SOM clustering, compared to the original dataset. The approach reduces bias and enhances autism diagnosis accuracy.

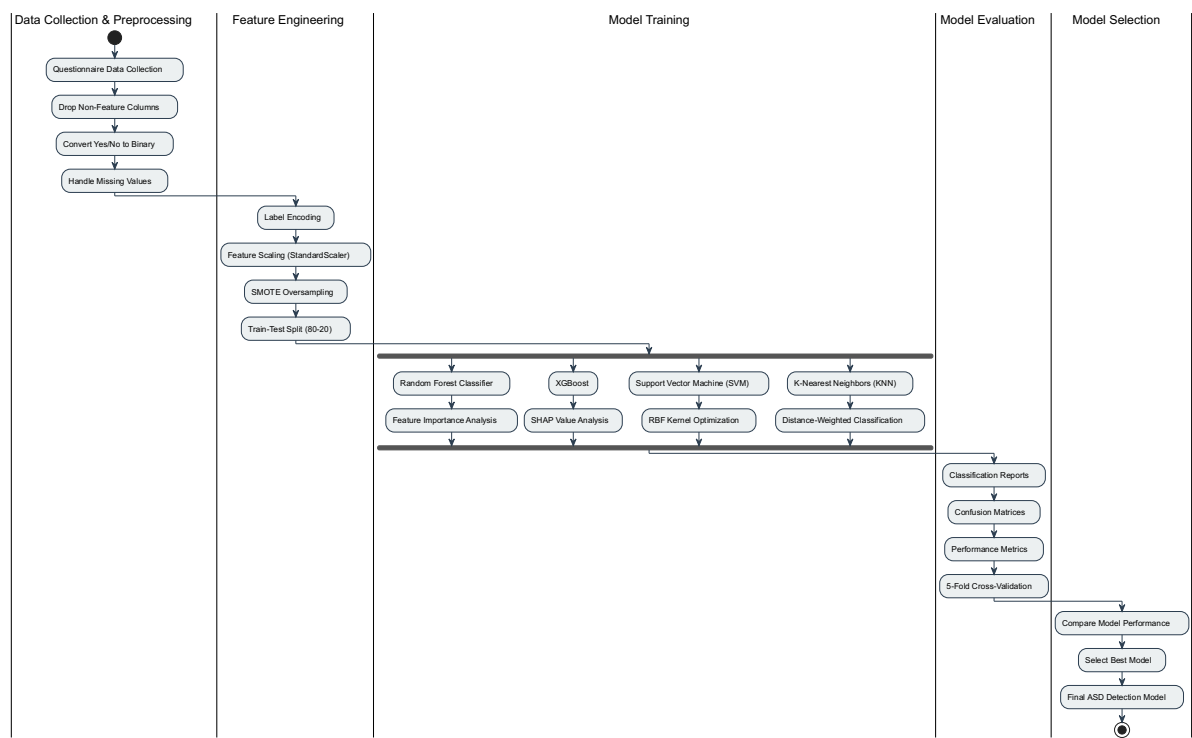
Various studies on autism spectrum disorder (ASD) detection utilize different machine learning (ML) and deep learning (DL) models, demonstrating strong performance in early diagnosis. **RF** achieved up to 97.58% accuracy, 100% sensitivity, and 96.66% specificity, outperforming other classifiers in several studies. The **RML** model, introduced by Thabtah & Peebles, achieved accuracies of 96.59% for adults, 93.70% for adolescents, and 94.20% for children, with sensitivity and specificity above 93%. **SVM** also showed strong results, with some studies reaching up to 100% accuracy. **CNN**, applied in deep learning-based ASD screening, achieved 97.95% accuracy and outperformed other models in sensitivity, specificity, and true positive rates. **ANN** were used for pattern detection in questionnaire data, with accuracies ranging from 90-100%. **AutoML** for feature selection and model training achieved high accuracy across age groups, including 96.1% for toddlers and 97.2% for adults. **DT** and **NB** were also used, with DT achieving high accuracy and NB showing results from 78-96%. These models were

evaluated using performance metrics such as accuracy, sensitivity, specificity, Area Under the Curve (AUC), and F1-score. They demonstrate significant potential in improving ASD detection accuracy, efficiency, and early diagnosis, especially when integrated with parent questionnaires and video screenings, enhancing early intervention and outcomes.

Sample collection methods and theory will be added.

Proposed method:

The proposed methodology presents a comprehensive machine learning framework for ASD detection through questionnaire-based assessment. Figure 1 illustrates the systematic approach, encompassing data preprocessing, feature engineering, model training, and evaluation phases.



The methodology initiates with data collection and preprocessing, where questionnaire responses are systematically processed to ensure data quality and consistency. Non-feature columns, such as timestamps and identifying information, are eliminated to maintain data relevance. A critical transformation involves converting categorical Yes/No responses to binary format (1/0), facilitating numerical analysis. Missing value imputation is implemented using a most-frequent strategy, ensuring dataset completeness without introducing significant bias.

In the feature engineering phase, the framework employs label encoding for categorical variables, followed by the application of Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance issues common in medical datasets. The data is strategically partitioned into training (80%) and testing (20%) sets, maintaining stratification to preserve class distribution.

The model training phase employs a parallel architecture integrating four distinct machine learning paradigms. The RF Classifier, an ensemble method, incorporates feature importance analysis to identify significant ASD predictors. XGBoost (XGB) implementation is enhanced with SHAP (SHapley Additive exPlanations) value analysis, providing interpretable outputs crucial for clinical applications. The Support Vector Machine (SVM) utilizes a Radial Basis Function (RBF) kernel with probability estimation capabilities, optimized through grid search for hyperparameter tuning. The K-Nearest Neighbors (KNN) algorithm implements a distance-weighted approach with $k=5$ neighbors, leveraging local feature space structure for classification. Model evaluation encompasses comprehensive performance assessment through multiple metrics. The framework employs classification reports, confusion matrices, and a 5-fold cross-validation strategy to ensure robust model validation while identifying optimization opportunities. This multi-metric approach provides thorough assessment of each model's capabilities across different data partitions, ensuring reliable performance evaluation.

The methodology concludes with a systematic model selection phase based on comparative performance analysis. Selection criteria incorporate multiple metrics including accuracy, precision, recall, and F1-score, while considering model interpretability and computational efficiency crucial for clinical deployment. The framework's strength lies in its diverse approach to classification, combining tree-based ensemble methods, kernel-based classification, and instance-based learning, thereby providing robust ASD detection capabilities while maintaining analytical transparency essential for clinical applications.

Results and discussion:

Our experimental evaluation encompassed four machine learning models RF, XGB, SVM, KNN for ASD detection, with particular emphasis on RF and XGB for feature importance analysis. The comprehensive performance metrics are presented in Tables 1-3.

Table 1: Model Performance Metrics

Metric	Random Forest	XGBoost	SVM	KNN
Accuracy	0.98	0.95	0.98	0.93
Macro Avg F1	0.98	0.95	0.98	0.93
Weighted Avg F1	0.98	0.95	0.98	0.93
CV Score ($\mu \pm \sigma$)	-	-	0.863 ± 0.061	0.863 ± 0.052

Table 2: Class-wise Performance Metrics

Class	Non-ASD (P/R/F1)	ASD (P/R/F1)
Random Forest	1.00 / 0.96 / 0.98	0.95 / 1.00 / 0.97
XGBoost	0.92 / 1.00 / 0.96	1.00 / 0.89 / 0.94
SVM	0.96 / 1.00 / 0.98	1.00 / 0.95 / 0.97
KNN	0.89 / 1.00 / 0.94	1.00 / 0.84 / 0.91

Table 3: Top 5 Important Features by Model

Rank	Random Forest (Feature / Importance)	XGBoost (Feature / Importance)
1	Turn towards person (0.112)	Meaningful communication (1.222)
2	Experience joy (0.107)	Imitation ability (0.924)
3	Imitation ability (0.096)	Mingling with kids (0.811)
4	Eye contact (0.081)	Potty trained (0.752)
5	Shoe identification (0.067)	Mouthing behavior (0.621)

The analysis revealed hierarchical model performance, with RF and SVM achieving superior accuracy (0.98), followed by XGB (0.95) and KNN (0.93). Cross-validation scores for SVM and KNN (0.863 ± 0.061 and ± 0.052 respectively) demonstrated robust generalization capabilities. Feature importance analysis, conducted through RF and XGB due to their interpretability characteristics, revealed crucial behavioral indicators. RF emphasized social responsiveness (turn towards person: 0.112) and emotional expression (experience joy: 0.107), while XGB highlighted communication abilities (meaningful communication: 1.222) and

social interaction skills (imitation ability: 0.924). The complementary insights from both models validate the significance of these behavioral markers in ASD detection. The integrated analysis validates our multi-model approach, with RF and SVM emerging as optimal choices for clinical applications. The consistent high performance across different algorithmic paradigms (ensemble, kernel-based, and instance-based) reinforces the methodology's reliability. Feature importance analysis through RF and XGB provides valuable insights for questionnaire refinement and diagnostic focus in clinical settings.

Conclusion:

This research advances the field of automated ASD detection through a comprehensive machine learning framework incorporating multiple algorithmic paradigms. Our study demonstrates the efficacy of ensemble methods (RF, XGB), kernel-based approaches (SVM), and instance-based learning (KNN) in achieving high-accuracy detection while maintaining clinical interpretability. The comparative analysis revealed superior performance metrics for Random Forest and SVM (98% accuracy), followed by XGB (95%) and KNN (93%). Cross-validation results (0.863 ± 0.061) validated the robustness of the models' generalization capabilities. The implementation of SMOTE for class imbalance mitigation, coupled with standardized feature scaling, contributed significantly to the framework's reliability. Feature importance analysis through Random Forest and XGBoost revealed crucial behavioral indicators, with communication abilities (importance: 1.222), social responsiveness (0.112), and imitation skills (0.924) emerging as primary predictors. This consistency across different modeling approaches validates these features' significance in ASD assessment, providing empirical support for clinical questionnaire optimization. The methodology's strength lies in its integrated approach to model development and evaluation, ensuring both predictive accuracy and clinical interpretability. Future research directions include exploring temporal dependencies through longitudinal studies and investigating advanced deep learning architectures while maintaining interpretability constraints essential for clinical applications. This study contributes significantly to automated medical diagnosis research, demonstrating the potential of multi-model machine learning approaches in supporting evidence-based clinical decision-making. The proposed framework establishes a foundation for developing more sophisticated ASD detection tools, potentially enhancing early intervention outcomes through improved diagnostic accuracy and interpretability.